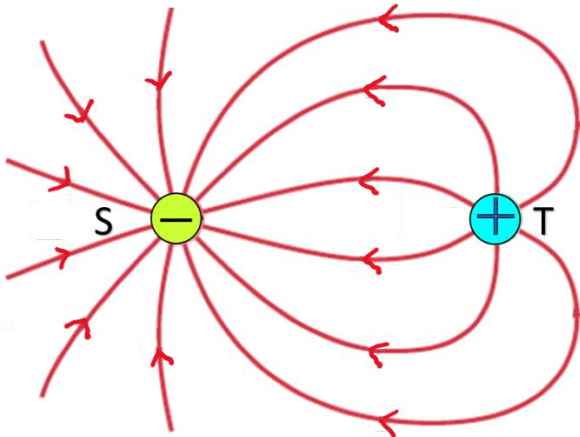
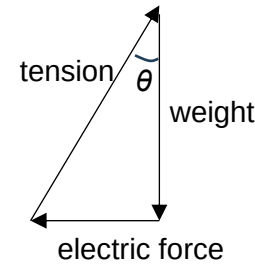
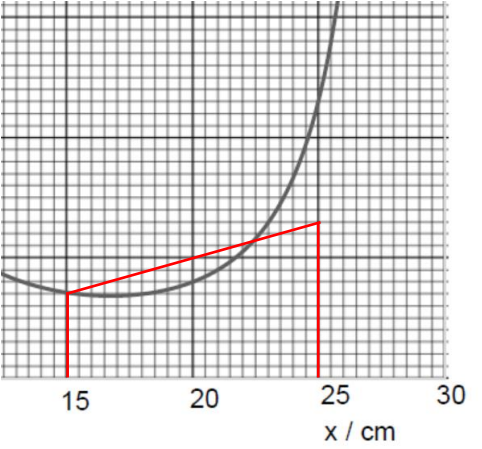


Q10		
(a)	Electric field strength at a point is the electric force per unit positive charge on a small test charge placed at the point.	B1
(b)(i)	 Correct direction. Correct ratio of lines (2:1). Correct asymmetry.	B1 B1 B1
(b)(ii)1	Consider the force-diagram of sphere T. The electric force must act horizontally to the left since S and T are align horizontally. The weight must act vertically down. Since T is in equilibrium under the effect of 3 forces, these three forces must form a closed right angle triangle as shown. $\text{Electric force} = \frac{1}{4\pi\epsilon_0} \frac{(2.4 \times 10^{-6})(1.2 \times 10^{-6})}{0.30^2}$ $= 0.287738 \text{ N}$ $\theta = \tan^{-1}\left(\frac{\text{electric force}}{\text{weight}}\right) = \tan^{-1}\left[\frac{0.287738}{0.036(9.81)}\right]$ $= 39^\circ$ 	M1 M1 A1
(b)(ii)2	Once the string is cut, the net force on sphere T will be the vector sum of the weight and electric force. $\Sigma F = ma$ $\sqrt{[(0.036)(9.81)]^2 + 0.287738^2} = (0.036)a$ $a = 13 \text{ m s}^{-2}$	B1 M1 A1

(b)(iii)	Area under the graph is change in potential. $\Delta V = \frac{1}{2}(0.10)[(0.14 + 0.26) \times 10^7]$ $= 2.0 \times 10^5 \text{ V}$ (accept +/- 10%) From 25 cm to 15 cm, the positive helium nucleus loses potential energy and gains kinetic energy.		B1 M1 M1 A1
(c)(i)	Weight of lithium ion = $mg = 6.941 \times 1.66 \times 10^{-27} \times 9.81 = 1.13 \times 10^{-25} \text{ N}$ Electric force on ion = $qE = 1.60 \times 10^{-19} \times \left(\frac{150}{0.020}\right) = 1.2 \times 10^{-15} \text{ N}$ The electric force is about 10^{10} times larger or 10 orders of magnitude larger than the gravitational force or weight.		M1 M1 A1
(c)(ii)	Time taken to travel through plates = $\frac{0.090}{3.0 \times 10^5} = 3.0 \times 10^{-7} \text{ s}$ Applying $s = ut + \frac{1}{2}at^2$ in the vertical direction, $y = \frac{1}{2} \left(\frac{1.2 \times 10^{-15}}{6.941 \times 1.66 \times 10^{-27}} \right) (3.0 \times 10^{-7})^2$ $y = 4.7 \text{ mm}$		M1 M1 A1